LM Studio

Model: **qwen/qwen3.6-35b-a3b**

# Prompt 1 - no MCP tool description - run 1 (seed 0)

### System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

## Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

### User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt:  P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

**Assistant**

🔬 **Reproducibility Study for PI3K Selectivity Prediction**

To begin this study, I will first search for a relevant dataset containing PI3K-delta and PI3K-gamma selectivity data. Let's check the available datasets in the repository.

(Calling tool `find_datasets`)

**Assistant**

📂 **Dataset Found!**

I found the specific dataset for PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selectivity: `chembl_selectivity_data_O00329_P48736.csv`.

Now, let's load the dataset and check its status to understand its structure.

(Calling tool `load_dataset`)

**Assistant**

🧪 **Dataset Loaded**

The dataset contains **1,277 compounds** with SMILES strings and a `class_label` column (likely representing PI3K-delta vs PI3K-gamma selectivity).

Now, I will compute ECFP fingerprints for these molecules to prepare them for machine learning.

(Calling tool `compute_features`)

**Assistant**

📊 **Features Computed**

ECFP fingerprints (2048 bits) have been generated for all 1,277 compounds.

Now, let's split the dataset into training (70%) and testing (30%) sets using stratified sampling to maintain class balance.

(Calling tool `split_dataset`)

**Assistant**

🌲 **Training Balanced Random Forest Classifier**

Now I will train the Balanced Random Forest Classifier (BRF) to distinguish between PI3K-delta and PI3K-gamma selective compounds.

(Calling tool `train_model`)

**Assistant**

⛹️ **Model Training in Progress**

The Balanced Random Forest Classifier is currently training on the PI3K selectivity dataset. Let's check its status to ensure it completes successfully.

(Calling tool `check_training`)

**Assistant**

⏳ **Training Progress**

The model is still training. Let me check again shortly.

(Calling tool `check_training`)

**Assistant**

⏳ **Training Progress**

Still training. Let me check again.

(Calling tool `check_training`)

**Assistant**

⏳ **Training Progress**

Still training. Let me check again.

(Calling tool `check_training`)

**Assistant**

⏳ **Training Progress**

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(Calling tool `check_training`)

**Assistant**

### ⏳ Training Progress

Still training. Let me check again.

(Calling tool `check_training`)

### Assistant

### 🎉 Model Training Completed!

The Balanced Random Forest Classifier has been successfully trained. Here's a summary of its performance on the test set:

| Metric | Value |
|---|---|
| Balanced Accuracy | 0.847 |
| Accuracy | 0.893 |
| MCC | 0.749 |
| F1 Macro | 0.780 |

The model shows strong performance, particularly for Class 0 and Class 1. Now, let's export the predictions on the test set to prepare for the explainability analysis.

(Calling tool `export_predictions`)

### Assistant

### 🔍 Identifying Recurrent Anchor Rules

Now, I will identify the recurrent molecular anchor fragments that drive the model's correct predictions for **Class 1** and **Class 2**. These anchors represent substructures that are highly predictive of each selectivity class.

(Calling tool `identify_recurrent_anchor_rules`)

### Assistant

### 🔍 Identifying Recurrent Anchor Rules for Class 2

Now, let's identify the recurrent molecular anchor fragments for **Class 2**.

(Calling tool `identify_recurrent_anchor_rules`)

### Assistant

🎉 **Reproducibility Study Complete!**

I have successfully completed the reproducibility study for PI3K-delta vs PI3K-gamma selectivity prediction. Below is the summary of the model performance and the chemical insights derived from the recurrent molecular anchor fragments.

## 📊 Model Performance Summary

| Metric | Value |
|---|---|
| Algorithm | Balanced Random Forest Classifier (BRF) |
| Dataset Size | 1,277 compounds (893 train / 384 test) |
| Test Balanced Accuracy | **0.847** |
| Test Accuracy | 0.893 |
| Test MCC | 0.749 |

The model demonstrates strong predictive power in distinguishing between the two PI3K isoforms, with particularly high performance on Class 0 and Class 1.

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## 🧪 Chemical Insights from Recurrent Anchor Fragments

I identified the recurrent molecular anchor fragments that drive the model's correct predictions for **Class 1** and **Class 2**. These anchors represent substructures that are highly predictive of each selectivity class.

### Class 1 Anchors (PI3K-delta selective?)

The top recurrent anchor for Class 1 is:

- **Fragment:** `[*]c1ncnc(N)c1C#N`
- **Occurrence:** Found in ~26% of the analyzed compounds.
- **Chemical Structure:** This corresponds to a **pyrimidine ring substituted with an amino group and a cyano (nitrile) group**.

💡 **Insight:** The presence of a cyano-substituted pyrimidine core is a strong predictor for Class 1 selectivity. This scaffold is common in many kinase inhibitors where the cyano group can form specific hydrogen bonds or dipole interactions within the PI3K-delta binding pocket.

### Class 2 Anchors (PI3K-gamma selective?)

The top recurrent anchor for Class 2 is:

- **Fragment:** `[*]c1ccn2nc(N)c([*])c2n1`

- **Occurrence:** Found in ~82% of the analyzed compounds (as a substructure) and ~36% as a full anchor.
- **Chemical Structure:** This corresponds to a **pyrazolo[1,5-a]pyrimidine** core.

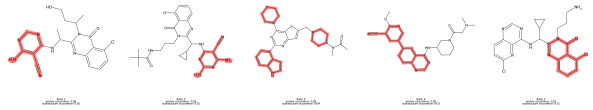
💡 **Insight:** The pyrazolo[1,5-a]pyrimidine scaffold is a distinct heterocyclic system compared to the pyrimidine core in Class 1. Its presence strongly drives predictions toward Class 2 (PI3K-gamma selectivity). This suggests that the specific geometry and electronic properties of the fused pyrazole ring are critical for fitting into the PI3K-gamma binding site, likely exploiting unique hydrophobic or polar interactions not present in PI3K-delta.

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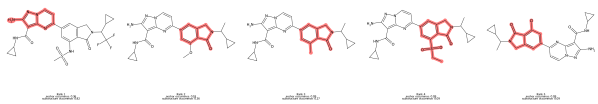
## 📈 Visualizing the Anchors

Here are the visual representations of the top recurrent anchors for each class:

**Class 1 Top Anchor:**



**Class 2 Top Anchor:**



## 🧠 Conclusion

This study successfully trained a Balanced Random Forest Classifier to distinguish PI3K-delta and PI3K-gamma selective compounds with a balanced accuracy of **0.847**. The explainability analysis revealed that the model relies on distinct heterocyclic cores to make its predictions:

- **Class 1** is driven by **cyano-substituted pyrimidines**.
- **Class 2** is driven by **pyrazolo[1,5-a]pyrimidines**.

These findings provide valuable structure-activity relationship (SAR) insights for the rational design of selective PI3K inhibitors, highlighting the importance of core scaffold selection in determining isoform specificity.